

MicroStation Distribution Service Connection Design Package

Drawing Production, Voltage Drop Analysis, Quantity Takeoff, and Field Verification
Planning for a Mock Commercial Customer Service Connection

Vivek Aruja

June 10, 2026

Abstract

This project documents a complete mock electrical distribution service connection design package, built to develop practical familiarity with MicroStation based utility design workflows and the supporting project control tasks typically associated with utility design and construction support roles. The package simulates a fictional commercial customer service connection involving an existing padmount transformer (TX-01), a proposed underground duct bank, a proposed pull box (PB-01), and a customer service entrance (SE-01), connected by an 85 m secondary route split into a 45 m segment and a 40 m segment.

The deliverable package includes a six-sheet MicroStation drawing set (cover sheet, existing conditions plan, proposed electrical civil plan, plan and profile drawing, single line diagram, and construction details), an Excel based voltage drop and quantity takeoff workbook, a project record tracker, and a field verification checklist. The assumed service is 600 V, three phase, supplying a 250 kVA commercial load, yielding a calculated current of approximately 240.56 A and a simplified voltage drop of approximately 7.69 V (1.28 % of nominal voltage) across the 85 m route.

The project is not based on real utility records, internal standards, or customer data, and is not for construction. It was created strictly as an academic portfolio exercise to demonstrate drawing production, calculation review, quantity estimation, and project documentation practices relevant to electrical distribution design.

Contents

1	Introduction	3
2	Project Scenario	4
2.1	Route and Equipment Summary	4
3	Design Package Overview	5
4	MicroStation Drawing Package	6
4.1	E-000 Cover Sheet	6
4.2	E-101 Existing Conditions Plan	7
4.3	E-102 Proposed Electrical Civil Plan	8
4.4	E-201 Plan and Profile Drawing	9
4.5	E-301 Single Line Diagram	10
4.6	E-401 Construction Details and Notes	11
5	Excel Calculations and Quantity Takeoff	12
5.1	Voltage Drop Calculation	12
5.2	Cable Pull Review	13
5.3	Quantity Takeoff	13
6	Project Records and Field Verification	14
6.1	Project Record Tracker	14
6.2	Field Verification Checklist	14
7	Discussion	15
8	Skills Demonstrated	15
9	Limitations	16
10	Future Work	16
11	Conclusion	17
	References	18

1 Introduction

Electrical distribution service connection design involves coordinating civil, electrical, and project control information into a single, consistent drawing and documentation package. A service connection links a utility's distribution system, typically through a padmount or pole mounted transformer, to a customer's service entrance, while accounting for route geometry, duct bank construction, voltage drop, and field conditions.

This report documents a mock electrical distribution service connection design package created to build practical familiarity with MicroStation based utility design workflows. The project simulates a fictional commercial customer service connection involving an existing padmount transformer, a proposed underground duct bank route, a proposed pull box, and a customer service entrance.

The purpose of the project was to practice the type of work commonly associated with utility design and construction support, including:

- Drawing production in MicroStation.
- Project record tracking.
- Voltage drop review.
- Quantity takeoff.
- Field verification planning.

The project followed the workflow below:

Scenario definition → drawing production → calculation review → quantity takeoff →
project record tracking → field verification planning

The project was created as an academic portfolio project and is not intended for construction use.

2 Project Scenario

The project scenario is a fictional commercial customer located at a mock site address. The customer requires a secondary service connection from an existing padmount transformer to a customer service entrance on the building.

2.1 Route and Equipment Summary

Table 1 summarizes the proposed route and equipment assumptions used throughout the design package.

Table 1: Project scenario summary.

Item	Value / Description
Existing transformer	TX-01 (existing padmount)
Proposed pull box	PB-01
Customer service entrance	SE-01
Route segment 1	TX-01 to PB-01, 45 m
Route segment 2	PB-01 to SE-01, 40 m
Total route length	85 m
Proposed duct bank	4 × 100 mm PVC ducts
Assumed service voltage	600 V, 3-phase
Assumed customer load	250 kVA

3 Design Package Overview

The project deliverables were organized into a small drawing and documentation package consistent with a typical utility design submission. The package includes MicroStation drawings, an Excel based calculation workbook, a project record tracker, and a field verification checklist.

Table 2 summarizes the drawing package.

Table 2: Drawing package summary.

Drawing No.	Drawing Title	Purpose
E-000	Cover Sheet and Drawing Index	Introduces the drawing package and drawing list
E-101	Existing Conditions Plan	Shows existing road, sidewalk, property, transformer, building, pole, and overhead line
E-102	Proposed Electrical Civil Plan	Shows proposed duct bank route from TX-01 to PB-01 to SE-01
E-201	Plan and Profile Drawing	Shows stationing, route length, duct depth, and field verification point
E-301	Single Line Diagram	Shows electrical source to load relationship
E-401	Construction Details and Notes	Shows duct bank detail, pull box detail, tagging, and quantity summary

4 MicroStation Drawing Package

4.1 E-000 Cover Sheet

The cover sheet provides the project title, drawing index, revision information, and title block. It establishes the drawing package structure and identifies the project as an academic portfolio project, not for construction.

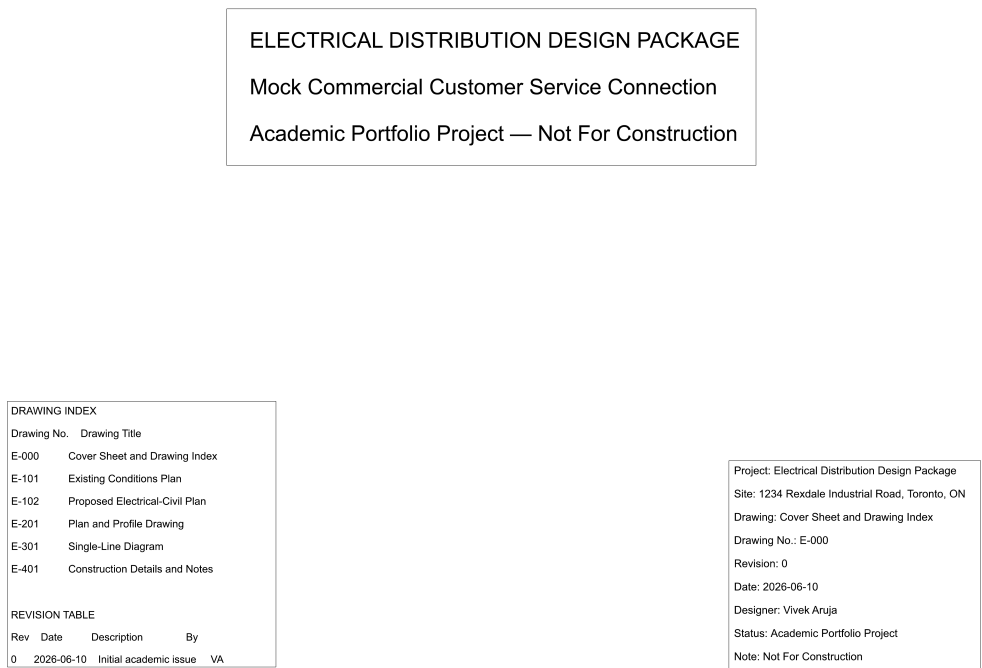
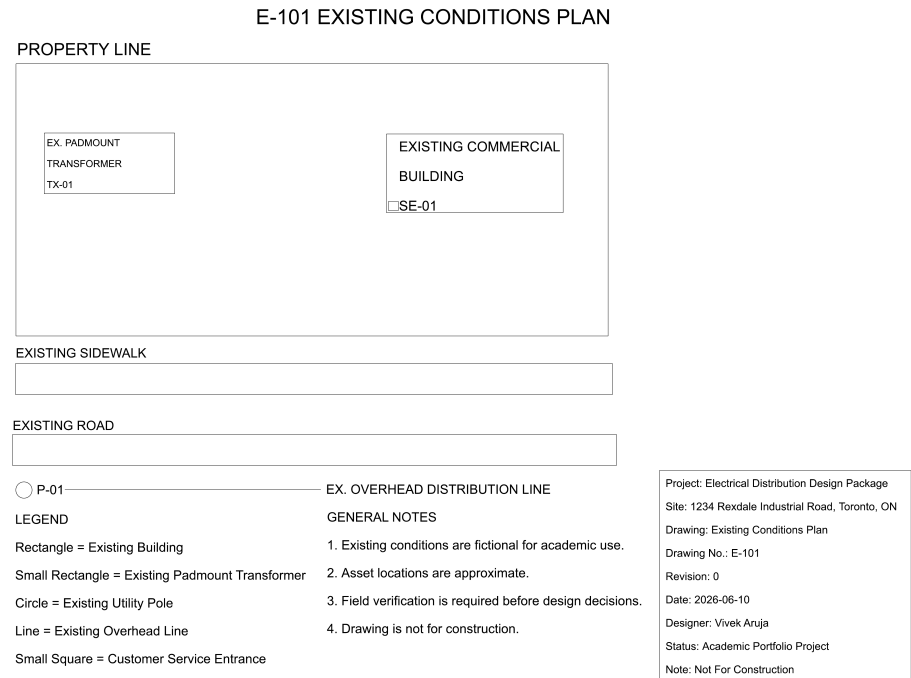


Figure 1: E-000 cover sheet and drawing index.

4.2 E-101 Existing Conditions Plan

The existing conditions plan shows the fictional site prior to proposed construction work. The drawing includes the property line, road, sidewalk, existing commercial building, existing padmount transformer, customer service entrance, utility pole, and overhead distribution line. Establishing existing conditions first is standard practice, since the proposed design must be drawn against a verified base.



4.3 E-102 Proposed Electrical Civil Plan

The proposed electrical civil plan adds the proposed underground duct bank route from TX-01 to PB-01 to SE-01. The route is divided into two segments: 45 m from TX-01 to PB-01 and 40 m from PB-01 to SE-01, giving a total assumed route length of 85 m. Splitting the route at PB-01 keeps each pull segment within a manageable length for cable installation.

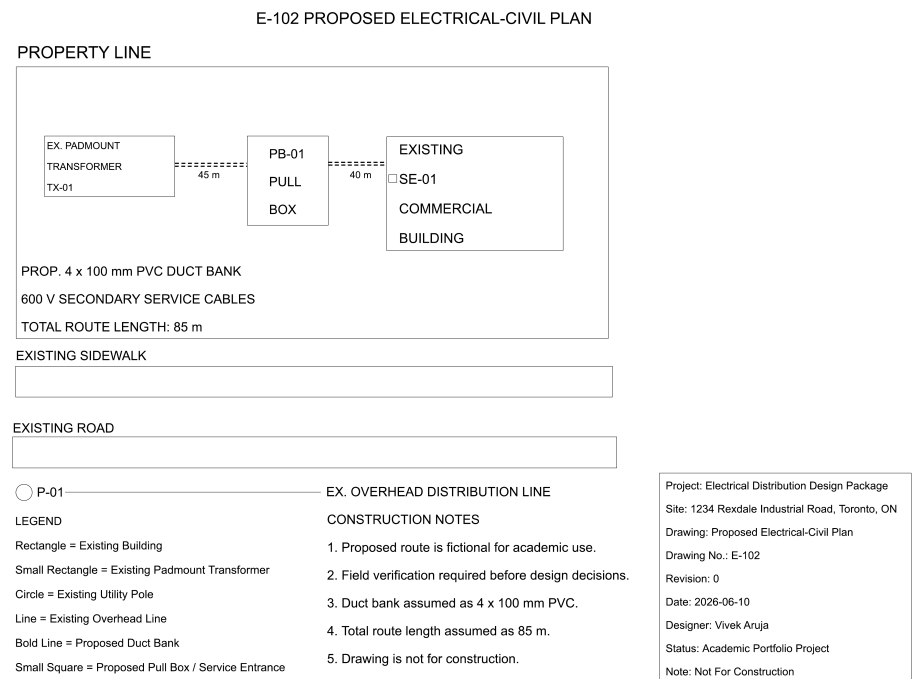


Figure 3: E-102 proposed electrical civil plan.

4.4 E-201 Plan and Profile Drawing

The plan and profile drawing expands the proposed route into a station-based layout. TX-01 is located at station 0+000, PB-01 at station 0+045, and SE-01 at station 0+085. The drawing includes an assumed existing grade elevation of 100.00 m and an assumed duct bank elevation of 99.10 m, representing a 0.90 m depth below grade.

An unknown utility crossing is included at station 0+030 to represent a field verification item, illustrating the importance of confirming utility conflicts before finalizing design or construction decisions.

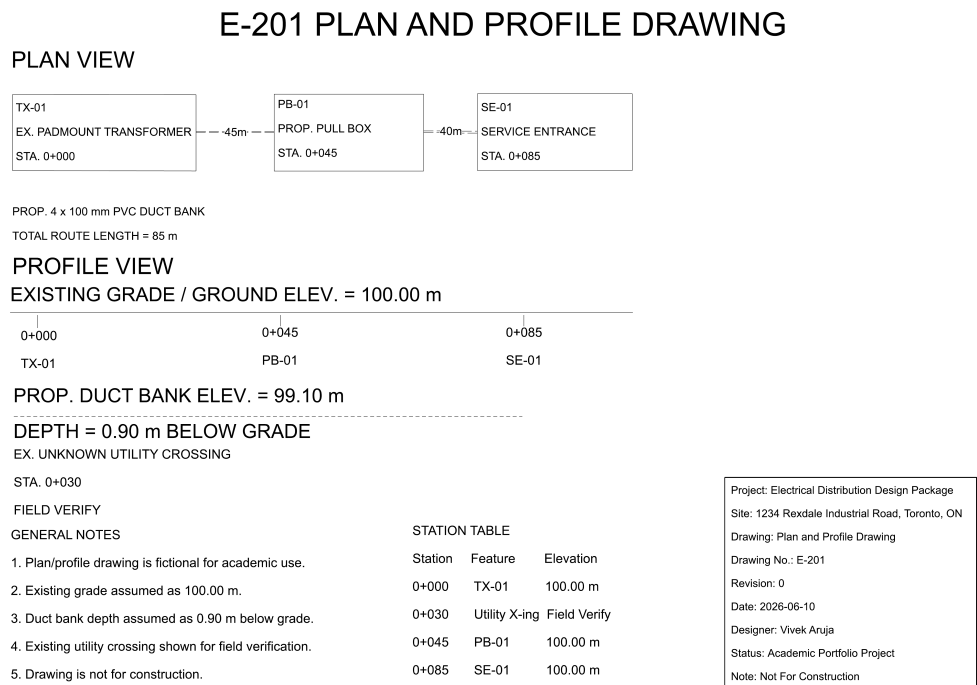


Figure 4: E-201 plan and profile drawing.

4.5 E-301 Single Line Diagram

The single line diagram shows the electrical relationship between the utility source, transformer, customer meter, main disconnect, customer distribution panel, and commercial load. The diagram assumes a 13.8 kV utility distribution source feeding a 13.8 kV/600 V transformer and a 600 V, three phase secondary service.

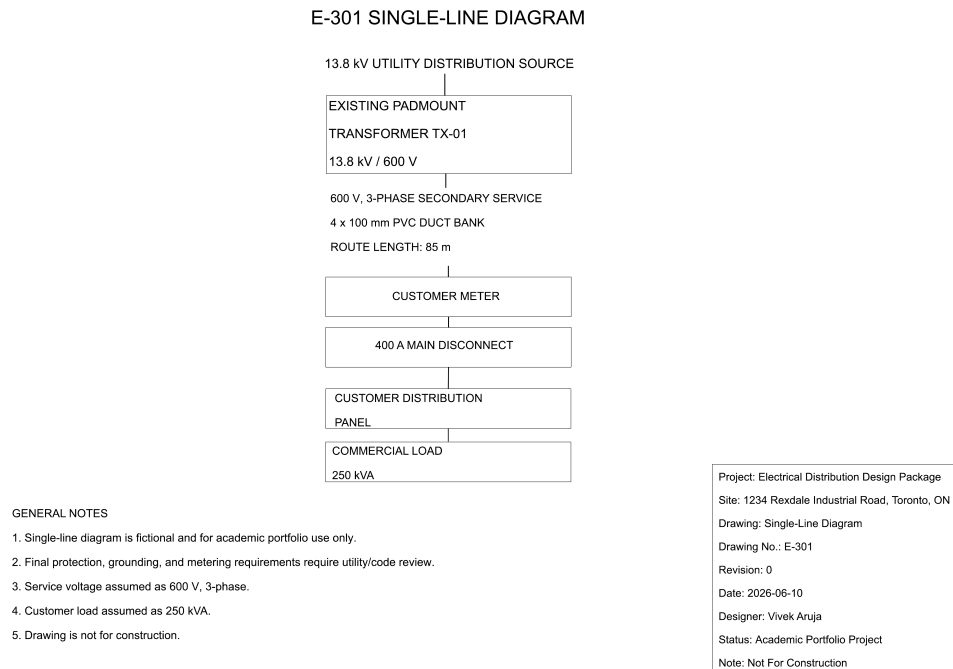


Figure 5: E-301 single-line diagram.

4.6 E-401 Construction Details and Notes

The construction details sheet includes a duct bank cross section, proposed pull box detail, cable/route tagging detail, material summary, and construction notes. The duct bank is represented as 4 × 100 mm PVC ducts arranged in a 2-over-2 configuration.

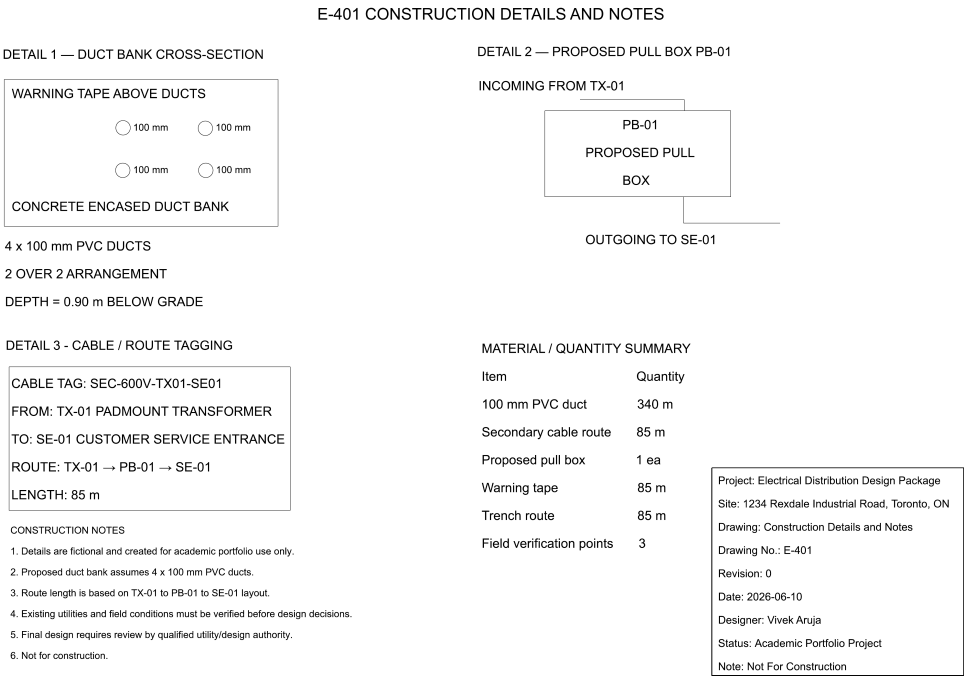


Figure 6: E-401 construction details and notes.

5 Excel Calculations and Quantity Takeoff

The Excel calculation workbook supports the drawing package by providing a simplified voltage drop check, cable pull review, and quantity takeoff. The workbook connects the drawing assumptions directly to calculation outputs and material quantities, mirroring how a real design package links drawings, calculations, and procurement.

5.1 Voltage Drop Calculation

The assumed service voltage is 600 V, three phase, and the assumed customer load is 250 kVA. The line current was calculated using the three phase power relationship:

$$I = \frac{S}{\sqrt{3} V} \quad (1)$$

Substituting the selected load and voltage:

$$I = \frac{250,000}{\sqrt{3}(600)} = 240.56 \text{ A} \quad (2)$$

A simplified voltage drop calculation was then performed using an assumed conductor resistance and the total route length:

$$V_{drop} = \sqrt{3} \times I \times R \times L \quad (3)$$

Substituting the calculated current, assumed resistance, and route length:

$$V_{drop} = \sqrt{3}(240.56)(0.000217)(85) = 7.69 \text{ V} \quad (4)$$

This corresponds to a voltage drop of approximately 1.28 % of the 600 V service voltage, well within a typical 3 % design guideline for secondary service runs.

Table 3: Voltage drop calculation summary.

Parameter	Value	Unit
Load	250	kVA
Voltage	600	V
Calculated current	240.56	A
Route length	85	m
Assumed conductor resistance	0.000217	Ω/m
Voltage drop	7.69	V
Voltage drop percent	1.28	%

5.2 Cable Pull Review

The proposed pull box PB-01 splits the route into two shorter pull segments. Segment 1 runs from TX-01 to PB-01 and is assumed to be 45 m; Segment 2 runs from PB-01 to SE-01 and is assumed to be 40 m, for a total route length of 85 m.

This is a simplified cable pull review intended to demonstrate the concept of splitting long runs at intermediate structures. A complete pulling tension and sidewall pressure calculation would require cable manufacturer data, detailed conduit bend geometry, friction coefficients, and construction specific assumptions, none of which were modelled in this academic exercise.

5.3 Quantity Takeoff

The quantity takeoff estimates the basic materials associated with the proposed route. Since the duct bank uses 4×100 mm PVC ducts over an 85 m route, the total PVC duct quantity is estimated as 340 m.

Table 4: Quantity takeoff summary.

Item	Quantity	Basis
100 mm PVC duct	340 m	4 ducts $\times$ 85 m
Secondary cable route	85 m	TX-01 to SE-01
Proposed pull box	1 ea	PB-01
Warning tape	85 m	Along trench route
Trench route	85 m	Total route length
Field verification points	3 ea	TX-01, PB-01, SE-01

6 Project Records and Field Verification

6.1 Project Record Tracker

A project record tracker was created to manage drawing numbers, drawing titles, revisions, PDF export status, and notes. This supports a project control workflow where drawings and related records are tracked consistently across the design package, similar to a drawing register used on a real utility project.

Project Schedule					
Phase	Deliverable	Start Date	Target Date	Status	Notes
1	Project folder setup	2026-06-10	2026-06-10	Complete	Folder structure, README, references, drawings, calculations
2	MicroStation drawing setup	2026-06-10	2026-06-10	Complete	Levels, DGN file, sheet models
3	E-000 Cover Sheet	2026-06-10	2026-06-10	Complete	Cover sheet and drawing index
4	E-101 Existing Conditions Plan	2026-06-10	2026-06-10	Complete	Site layout and existing assets
5	E-102 Proposed Electrical-Civil Plan	2026-06-10	2026-06-10	Complete	Proposed duct bank, pull box, route lengths
6	E-301 Single-Line Diagram	2026-06-10	2026-06-10	Complete	Electrical source-to-load diagram
7	E-201 Plan/Profile Drawing	2026-06-10	2026-06-10	Complete	Stationing, depth, utility crossing
8	E-401 Construction Details	2026-06-10	2026-06-10	Complete	Duct bank detail, pull box detail, quantity summary
9	Excel calculations workbook	2026-06-10	2026-06-10	Complete	Voltage drop, cable pull review, quantity takeoff
10	Field verification checklist	2026-06-11	2026-06-11	Not Started	Field measurement checklist
11	README and GitHub cleanup	2026-06-11	2026-06-11	Not Started	Final documentation and upload

Figure 7: Project record tracker.

6.2 Field Verification Checklist

A field verification checklist was created to identify items that would need confirmation before design or construction decisions are finalized. The checklist includes:

- Transformer location.
- Service entrance location.
- Pull box location.
- Route measurements.
- Existing utility conflicts.
- Road and sidewalk constraints.

Field Verification Checklist							
Check ID	Field Item	Location / Asset	Verification Required	Method	Status	Notes	Related Drawing
FV-001	Confirm transformer location	TX-01	Verify padmount transformer position	Field observation	Not Started	Confirm asset location before design decisions	E-101 / E-102
FV-002	Confirm service entrance location	SE-01	Verify customer service entrance on building	Field observation	Not Started	Confirm exact wall location and access	E-101 / E-102
FV-003	Confirm proposed pull box location	PB-01	Verify proposed location is accessible	Field measurement	Not Started	Check clearance and accessibility	E-102 / E-201
FV-004	Measure TX-01 to PB-01 route	Segment 1	Confirm 45 m route length	Tape / measuring wheel	Not Started	Compare to drawing assumption	E-102 / E-201
FV-005	Measure PB-01 to SE-01 route	Segment 2	Confirm 40 m route length	Tape / measuring wheel	Not Started	Compare to drawing assumption	E-102 / E-201
FV-006	Check existing utility crossing	STA. 0+030	Verify unknown utility conflict	Utility locate / field review	Not Started	Marked as field verify in profile	E-201
FV-007	Check road and sidewalk constraints	Existing road/sidewalk	Confirm constructability near public area	Field observation	Not Started	Check restoration/access concerns	E-101 / E-102
FV-008	Confirm trench path clearance	Duct bank route	Verify proposed trench route has no obvious obstruction	Field observation	Not Started	Check landscaping, access, structures	E-102 / E-201
FV-009	Confirm title block/project records	All drawings	Check drawing numbers and revisions	Record review	Complete	Revision 0 academic issue	All drawings
FV-010	Confirm not-for-construction note	All documents	Verify academic disclaimer is included	Record review	Complete	Included on drawings/checklists	All drawings

Figure 8: Field verification checklist.

7 Discussion

The completed package illustrates how drawing production, calculation review, and project documentation connect within a typical utility distribution design workflow. The existing conditions plan establishes the baseline site, the proposed electrical civil plan and plan/profile drawing define the route geometry and depth, the single line diagram confirms the electrical relationship between source and load, and the construction details sheet ties the route back to a material quantity summary.

The voltage drop result of 1.28 % over an 85 m run confirms that, under the simplified assumptions used, the secondary service does not require any special derating or oversized conductor to remain within typical voltage drop guidelines. The inclusion of an unverified utility crossing at station 0+030 was a deliberate design choice, intended to demonstrate why field verification is treated as a distinct deliverable rather than an afterthought: design assumptions made at the desk must be confirmed against actual site conditions before construction proceeds.

Splitting the route at PB-01 also reflects standard cable pulling practice, where long secondary runs are broken into shorter segments to remain within reasonable pulling tension and to provide an accessible point for future maintenance or fault location.

8 Skills Demonstrated

This project demonstrates the following skills relevant to utility design and construction support:

- Creating and organizing a MicroStation drawing package.
- Setting up drawing sheets, title blocks, levels, and drawing records.
- Preparing existing conditions and proposed electrical civil plan drawings.
- Creating a basic electrical single-line diagram.
- Creating a plan/profile drawing with stationing, duct depth, and a field verification point.
- Performing a simplified voltage drop calculation in Excel.
- Preparing a cable pull review and quantity takeoff tables.
- Managing project records using Excel.

9 Limitations

This project uses simplified assumptions for academic and portfolio purposes, and several limitations should be noted.

First, the drawings are not based on real utility records, internal design standards, confidential project information, or actual customer data; all dimensions, loads, and site conditions are fictional.

Second, the voltage drop calculation is preliminary and uses an assumed, generic conductor resistance rather than a manufacturer specific cable datasheet value; it does not replace a detailed design review.

Third, cable pulling tension, sidewall pressure, conduit fill, protection coordination, grounding, metering, civil restoration, and construction sequencing were not modelled in detail and would require review by qualified professionals and the relevant utility or design authority.

Fourth, the unknown utility crossing shown at station 0+030 is illustrative only and does not represent an actual field verified condition.

This project is not for construction.

10 Future Work

Future iterations of this exercise could be expanded in several directions. A detailed conduit fill and cable ampacity check could be added to confirm that the selected 100 mm duct size and conductor are appropriately matched to the 250 kVA load. A full cable pulling tension and sidewall pressure calculation, incorporating bend geometry and manufacturer specific friction coefficients, would strengthen the cable pull review beyond its current simplified form.

The project record tracker could also be expanded into a small relational database or linked workbook to better simulate a multi drawing, multi revision project environment, and the field verification checklist could be tied to a mock site visit report to close the loop between design assumptions and confirmed field conditions.

11 Conclusion

The MicroStation Distribution Service Connection Design Package demonstrates a complete mock workflow for a utility style service connection project. The project includes drawing production, proposed route design, single line representation, plan/profile documentation, construction details, Excel based calculations, project record tracking, and field verification planning.

The completed package shows how engineering drawings, calculations, and records connect in a practical utility design workflow, with the route design, voltage drop result, and quantity takeoff all traceable back to the same set of project assumptions. It was built to strengthen familiarity with MicroStation, Excel, and project documentation tasks relevant to electrical distribution design and construction support roles.

References

1. Ontario Electrical Safety Code, 28th Edition, Electrical Safety Authority.
2. IEEE Std 141, *Recommended Practice for Electric Power Distribution for Industrial Plants*.
3. Bentley Systems, *MicroStation User Guide*.
4. Toronto Hydro, *General Public Distribution Standards* (general/public information only, no confidential or internal standards used).